

Flatcar Orchestration & Container Runtimes

Source section: `orchestrate/` (renamed from `container-runtimes/` + parts of `setup/clusters/` in the refactored branch). Covers Docker, Kubernetes, and cluster configuration.

Containers (`orchestrate/containers/`)

Docker and containerd ship as built-in sysexst images enabled by default since Flatcar 3794.0.0.

Docker customization via `/etc/docker/daemon.json` :

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/docker/daemon.json
      contents:
        inline: |
          {
            "log-driver": "journald",
            "storage-driver": "overlay2"
          }
```

Custom Docker/containerd version: Disable the built-in sysexts (see [flatcar-provisioning](#)) and place custom binaries from `sysexst-bakery` or `/opt/bin/` .

Registry authentication: Credentials in `~/.docker/config.json` or `/etc/docker/config.json` for system-wide access.

Switching to unified cgroups (cgroupv2): Covered in `orchestrate/containers/switching-to-unified-cgroups.md` — relevant for Kubernetes nodes.

Incus (LXC/VM manager): Opt-in sysexst (`incus`) since Flatcar 4285.0.0. See `orchestrate/containers/incus.md` .

Kubernetes (`orchestrate/kubernetes/`)

Flatcar is a first-class Kubernetes node OS. Kubernetes is not bundled but deploys cleanly via `kubeadm` + sysexst images from `sysexst-bakery` .

Compatibility (as of 2026)

Kubernetes	Alpha	Beta	Stable	LTS (2024)
1.31–1.33	Known working	Known working	Known working	Known working
1.34–1.36	Tested	Tested	Tested	Tested

Tested CNIs: **Cilium**, **Flannel**, **Calico**

Known issue: Flannel >0.17.0 incompatible with enforced SELinux.

kubeadm + sysexth (recommended)

Use the Kubernetes sysexth from `sysexth-bakery` with `systemd-sysupdate` for in-place Kubernetes version upgrades.

Butane snippet (control plane):

```
variant: flatcar
version: 1.0.0
storage:
  links:
    - target: /opt/extensions/kubernetes/kubernetes-v1.33.2-x86-64.raw
      path: /etc/extensions/kubernetes.raw
  files:
    - path: /etc/sysupdate/kubernetes.d/kubernetes-v1.33.conf
      contents:
        source:
          https://extensions.flatcar.org/extensions/kubernetes/kubernetes-v1.33.conf
    - path: /opt/extensions/kubernetes/kubernetes-v1.33.2-x86-64.raw
      contents:
        source: https://extensions.flatcar.org/extensions/kubernetes-
v1.33.2-x86-64.raw
    - path: /etc/hostname
      contents:
        inline: "flatcar-node1"
```

Reboot coordination: Use [FLUO](#) (preferred) or [Kured](#) instead of locksmithd for Kubernetes nodes.

High Availability Kubernetes

- External load balancer for the API server
- etcd: stacked (on control plane nodes) or external

- FLUO drains and reboots nodes safely during OS updates

Clusters (`orchestrate/clusters/`)

Topic	Notes
Cluster architectures	Single-master, multi-master HA, worker-only roles
Discovery	etcd-based cluster discovery via a discovery URL
ECS	Flatcar nodes can register with Amazon ECS